

An NLP-Based Complaint Classification System for Barangay Reporting with Human-in-the-Loop Learning

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By

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Abstract

Barangay offices in the Philippines handle a high volume of citizen complaints related to public services, safety, and community concerns. However, existing complaint management systems still rely heavily on manual validation, leading to delays, inefficiencies, and inconsistent processing. Although e-governance platforms have improved digital submission and record-keeping, they often lack intelligent mechanisms for automatically classifying and filtering complaint reports.

This study proposes a Barangay Citizen Complaint System that integrates Natural Language Processing (NLP), Term Frequency–Inverse Document Frequency (TF-IDF), and Support Vector Machine (SVM) to enable automated complaint classification. The system processes multilingual inputs in Bisaya, Filipino, and English and analyzes textual content to improve classification accuracy. A Human-in-the-Loop (HITL) mechanism is incorporated to allow continuous model improvement through administrative feedback. In addition, a state-transition workflow is implemented to track complaint progression from submission to resolution.

The system classifies complaints into valid, suspicious, and spam categories, aiming to reduce manual workload and improve efficiency, consistency, and reliability in barangay complaint handling systems.

Keywords: Natural Language Processing, TF-IDF, Support Vector Machine, Human-in-the-Loop, Barangay Governance

Chapter 1

Introduction

1.1 Background of the Study

Citizen complaint management is an essential function of barangay governance in the Philippines, where residents report concerns such as public safety issues, infrastructure problems, and community disturbances. Despite its importance, many barangay offices continue to rely on manual validation processes in handling complaints, resulting in delays, inconsistencies in evaluation, and inefficiencies in overall service delivery.

Studies such as Laurena et al. (2022) and Aliling et al. (2025) highlight that the adoption of digital barangay systems can significantly improve administrative efficiency and transparency. However, existing systems primarily focus on digitizing complaint submission and record management rather than providing intelligent mechanisms for automatically classifying complaints or filtering invalid submissions. Platforms such as the Barangay Information Profiling System (BIPS) by Lacasandile et al. (2021) and AidPack by Laurena et al. (2022) demonstrate improvements in workflow digitization, but they still lack automated classification capabilities, adaptive learning mechanisms, and structured lifecycle tracking.

To address these limitations, Natural Language Processing (NLP) techniques have been widely applied in multilingual text analysis. Studies by Santiago (2023) and Wang (2021) show that NLP can effectively process Filipino, Tagalog, Bisaya, and code-switched text in real-world applications. In addition, Term Frequency–Inverse Document Frequency (TF-IDF) improves feature representation for text classification, while Support Vector Machine (SVM) has demonstrated strong performance in complaint classification tasks (Bojjireddy et al., 2021; Lin, 2022).

However, most existing systems still rely heavily on manual validation and lack mechanisms for continuous learning and improvement. There is also limited integration of Human-in-the-Loop (HITL) approaches that allow systems to adapt based on administrative feedback. Additionally, structured workflow tracking of complaint lifecycles remains underexplored in many existing implementations.

Therefore, there is a need for an integrated framework that combines NLP-based

text classification, supported by adaptive learning through Human-in-the-Loop (HITL) and a structured state-transition workflow, to improve the accuracy, efficiency, and reliability of barangay complaint management systems.

1.2 Statement of the Problem

ThThis study aims to develop an automated barangay complaint classification system using Natural Language Processing (NLP) and machine learning to improve the accuracy, efficiency, and reliability of complaint processing.

Specifically, the study seeks to answer the following questions:

1. How can a web-based barangay complaint system be developed to support efficient submission, processing, and lifecycle tracking of citizen complaints?
2. How can Natural Language Processing (NLP) techniques using TF-IDF and Support Vector Machine (SVM) classify complaints into valid, suspicious, and spam categories?
3. How can a Human-in-the-Loop (HITL) mechanism be integrated to allow continuous improvement of the classification model through administrative feedback?
4. How effective is the proposed system in terms of accuracy, precision, recall, and F1-score in classifying complaints compared to baseline or traditional methods?

1.3 Objectives of the Study

This study aims to develop and evaluate an automated barangay complaint classification system using Natural Language Processing (NLP) and machine learning.

Specifically, the study aims to:

1. Develop a web-based barangay complaint system that streamlines the submission, processing, and lifecycle tracking of citizen complaints.
2. Evaluate the effectiveness of TF-IDF and Support Vector Machine (SVM) in classifying multilingual complaint text (Bisaya, Filipino, and English) into valid, suspicious, and spam categories.
3. Implement a Human-in-the-Loop (HITL) mechanism that enables continuous improvement of the classification model through administrative feedback.
4. Evaluate the performance of the proposed system in terms of accuracy, precision, recall, F1-score, and reduction in manual validation workload compared to traditional methods.

1.4 Significance of the Study

This study contributes to the advancement of intelligent e-governance systems by improving barangay-level complaint management through automated text classification. It addresses inefficiencies in manual validation and supports more efficient and data-driven public service processes.

From a technical perspective, this research contributes to Natural Language Processing (NLP) and machine learning classification by applying these methods in a multilingual context involving Bisaya, Filipino, and English. It also extends existing work by integrating a Human-in-the-Loop (HITL) mechanism, enabling continuous model improvement through administrative feedback.

The study introduces a framework that combines NLP-based text classification with a state-transition workflow to ensure structured tracking of complaint lifecycles. This enhances system accountability, transparency, and overall process management.

For stakeholders, the system provides practical benefits to barangay officials by reducing manual workload and improving consistency in complaint processing, while also enhancing citizen experience through faster and more transparent complaint handling.

Overall, the study bridges the gap between traditional e-governance systems and adaptive intelligent solutions, offering a scalable approach that can be adopted by other local government units to improve administrative efficiency and transparency.

1.5 Scope and Limitations

Scope

This study focuses on the development of a web-based barangay complaint management system that utilizes Natural Language Processing (NLP), Term Frequency–Inverse Document Frequency (TF-IDF), and Support Vector Machine (SVM) for automated text classification. The system processes multilingual complaint inputs in Bisaya, Filipino, and English and classifies them into valid, suspicious, or spam categories.

The system incorporates a Human-in-the-Loop (HITL) feedback mechanism that allows model retraining based on administrative corrections. It also includes a state-transition workflow for tracking the lifecycle of complaints from submission to resolution.

Limitations

This study is limited to simulated or collected barangay complaint datasets and does not include full-scale real-time deployment across multiple barangays. The system is restricted to text-based complaints and does not support multimedia inputs such as images, videos, or audio.

Additionally, the system does not include advanced user identity verification and cannot fully guarantee the authenticity of anonymous submissions. The model's performance may vary depending on dataset size, data quality, and the level of code-switching

in multilingual inputs. Lastly, the system functions only as a decision-support tool and does not replace the final judgment of barangay officials.

Chapter 2

Review of Related Works

This chapter presents a review of existing literature organized thematically to identify the gaps that this study addresses. The themes cover barangay complaint management systems, NLP and machine learning for text classification, Human-in-the-Loop and adaptive learning systems, and workflow and lifecycle management.

2.1 Barangay Complaint Management Systems

Citizen complaint management is a core function of barangay governance in the Philippines, where residents report concerns such as public safety issues, environmental problems, infrastructure issues, and community-related incidents. As the frontline of local governance, barangays are expected to respond quickly, transparently, and efficiently to citizen concerns. However, complaint handling in most barangays is still largely conducted through manual processes that rely on physical records, face-to-face reporting, and administrative validation by barangay personnel.

This manual system presents several challenges. It often leads to delayed processing due to the time required for sorting, reviewing, and validating complaints. It also introduces inconsistencies in decision-making since validation depends on human judgment, which may vary across personnel. In addition, manual handling increases the risk of errors when dealing with large volumes of complaints, particularly during peak periods. The absence of a standardized workflow further makes it difficult to track the full lifecycle of complaints from submission to resolution.

In response to these challenges, studies such as Laurena et al. (2022) and Alil-ing et al. (2025) show that digital barangay systems improve efficiency, transparency, and data management. Platforms like AidPack and E-BIMS demonstrate that web-based systems can streamline complaint submission and centralize records. However, these systems remain primarily focused on digitization rather than intelligent decision-making, as they do not automatically classify complaints or filter invalid submissions. Most existing systems lack automated mechanisms for complaint classification and do not incorporate adaptive learning or structured workflow intelligence, meaning complaints still require manual validation and tracking by barangay staff.

2.2 NLP and Machine Learning for Text Classification

Natural Language Processing (NLP) is a branch of artificial intelligence that enables machines to understand and process human language. It is widely used in analyzing unstructured text data such as user-generated reports and complaints. In multilingual environments such as the Philippines, NLP is particularly useful in processing Filipino, English, and Bisaya text, including code-switched communication (Santiago, 2023; Wang, 2021).

To transform textual data into machine-readable format, Term Frequency–Inverse Document Frequency (TF-IDF) is used as a feature extraction technique. It assigns higher weight to important words while reducing the influence of common terms, enabling meaningful representation of complaint content for machine learning models. Bojjireddy et al. (2021) found that TF-IDF significantly improves classification performance when combined with supervised learning methods.

Support Vector Machine (SVM) has demonstrated strong performance as a classification model for text-based tasks. It performs well in high-dimensional spaces and sparse datasets, making it suitable for complaint classification. Studies by Lin (2022) and Bojjireddy et al. (2021) confirm the strong performance of SVM in classification problems involving textual data. Li et al. (2025) further demonstrated the effectiveness of combining NLP and machine learning in an intelligent complaint classification system for medical settings, achieving high accuracy in categorizing patient complaints.

Despite the availability of these techniques, many existing systems still face limitations in handling multilingual and code-switched inputs. Furthermore, most systems do not incorporate adaptive mechanisms that allow models to improve based on real-world administrative feedback.

2.3 Human-in-the-Loop and Adaptive Learning Systems

Many existing text classification systems operate as static models that do not improve after initial deployment. This is a significant limitation in real-world environments where language patterns evolve over time. Human-in-the-Loop (HITL) is an approach that addresses this by integrating human expertise into the machine learning lifecycle to improve model performance and reliability.

Amershi et al. (2014) demonstrated that incorporating human feedback into interactive machine learning significantly improves system accuracy and user trust. HITL is commonly implemented through Active Learning, where the model selectively queries human experts for labels on uncertain or ambiguous data, thereby improving accuracy while reducing the need for large labeled datasets. Settles (2009) confirmed that Active Learning is particularly effective in dynamic environments where data patterns continuously evolve, such as changes in language use or emerging spam behaviors.

In the context of barangay complaint systems, most existing platforms do not

incorporate any form of adaptive learning. Administrative corrections are rarely captured and used to retrain models, resulting in static systems that cannot adapt to new complaint patterns or local linguistic variations. This study addresses this gap by integrating a HITL feedback loop that records administrative corrections as ground truth data for periodic model retraining.

2.4 Workflow and Lifecycle Management

Beyond classification, effective complaint management requires structured tracking of each complaint from submission to resolution. Most existing barangay systems lack a standardized workflow, making it difficult to monitor complaint status, assign responsibilities, or ensure timely resolution.

State transition theory provides a formal foundation for modeling structured workflows. It defines the lifecycle of a process as a series of discrete states and transitions triggered by specific events or actions. In complaint management, this translates to clearly defined states such as Pending, In Progress, Resolved, and Closed, each representing a stage in the complaint lifecycle.

Studies on e-governance systems, including Lacasandile et al. (2021), highlight the importance of structured dashboards and lifecycle tracking for improving administrative decision-making and accountability. However, these systems still lack integration with intelligent classification mechanisms, resulting in workflows that are structured but not automated. This study addresses that gap by combining automated classification with a state-transition workflow to provide both intelligent filtering and structured tracking.

2.5 Theoretical Framework

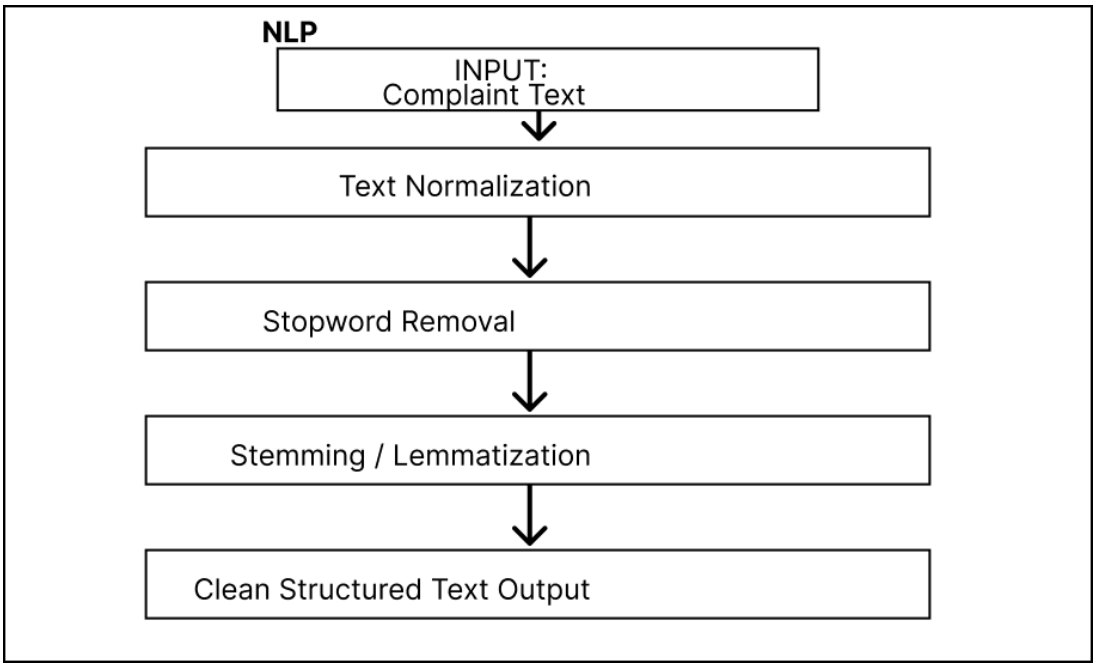
The theoretical framework of this study is anchored on six key theories and models: Natural Language Processing (NLP), Text Classification Theory, Term Frequency–Inverse Document Frequency (TF-IDF), Support Vector Machine (SVM), Human-in-the-Loop (HITL) and Active Learning, and State Transition Theory. Each theory is discussed individually below to establish the academic foundation of the proposed system.

2.5.1 Natural Language Processing (NLP)

Natural Language Processing is a field of artificial intelligence that enables computers to transform unstructured human language into structured and machine-readable data. In this study, NLP serves as the foundational layer that processes raw complaint text submitted by barangay residents, allowing it to be analyzed by machine learning models. As demonstrated by Santiago (2023), NLP techniques such as sentiment analysis and topic modeling can effectively analyze Filipino-language text, while Wang (2021) showed that multilingual models can process Tagalog and mixed-language inputs.

NLP operates through preprocessing steps such as text normalization, tokenization, and stopword removal. Further refinement is achieved through stemming or lemmatiza-

tion, which reduces words to their base form. These steps are particularly important in multilingual environments such as Davao City, where complaints may contain Filipino, Bisaya, and English. After preprocessing, NLP enables the transformation of text into structured representations for feature extraction and classification.



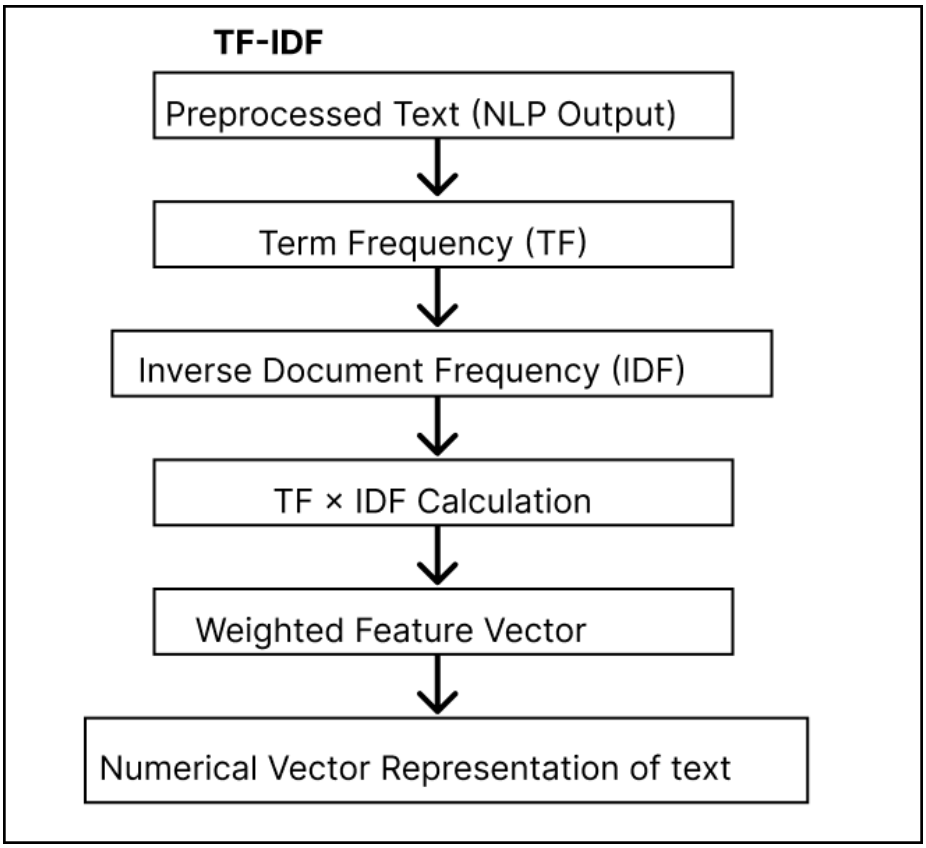
2.5.2 Text Classification Theory

Text classification theory explains the process of assigning predefined categories to textual data. It operates by learning patterns from labeled training data and applying these learned patterns to unseen inputs. This involves transforming text into numerical representations and using decision-making models to determine the most appropriate category, ensuring consistent and automated evaluation.

In this study, text classification theory underpins the entire complaint categorization process. The system learns from labeled complaint samples to distinguish between valid, suspicious, and spam complaints, enabling consistent and automated classification without relying on manual judgment.

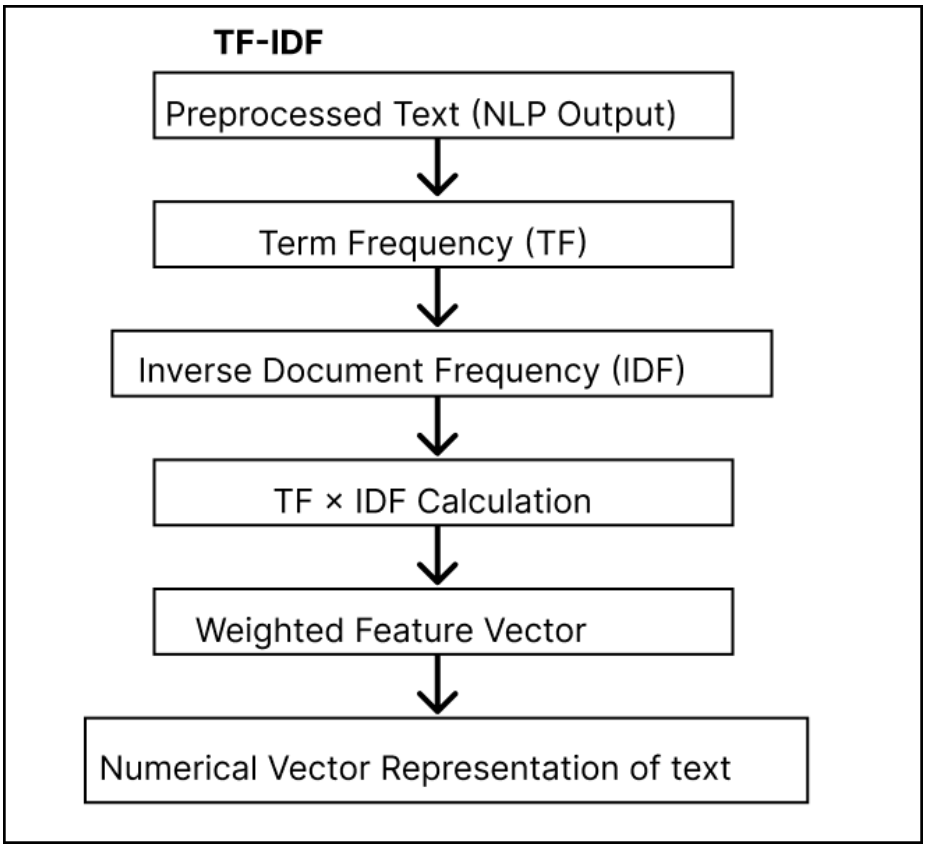
2.5.3 Term Frequency–Inverse Document Frequency (TF-IDF)

TF-IDF is a feature extraction technique that converts textual data into numerical representations based on word importance. It assigns higher weight to words that are significant within a specific document while reducing the influence of common terms that appear across many documents. According to Bojjireddy et al. (2021), TF-IDF significantly improves classification performance when combined with machine learning algorithms. In this study, TF-IDF transforms multilingual complaint text into structured numerical vectors, enabling the SVM classifier to process the data effectively.



2.5.4 Support Vector Machine (SVM)

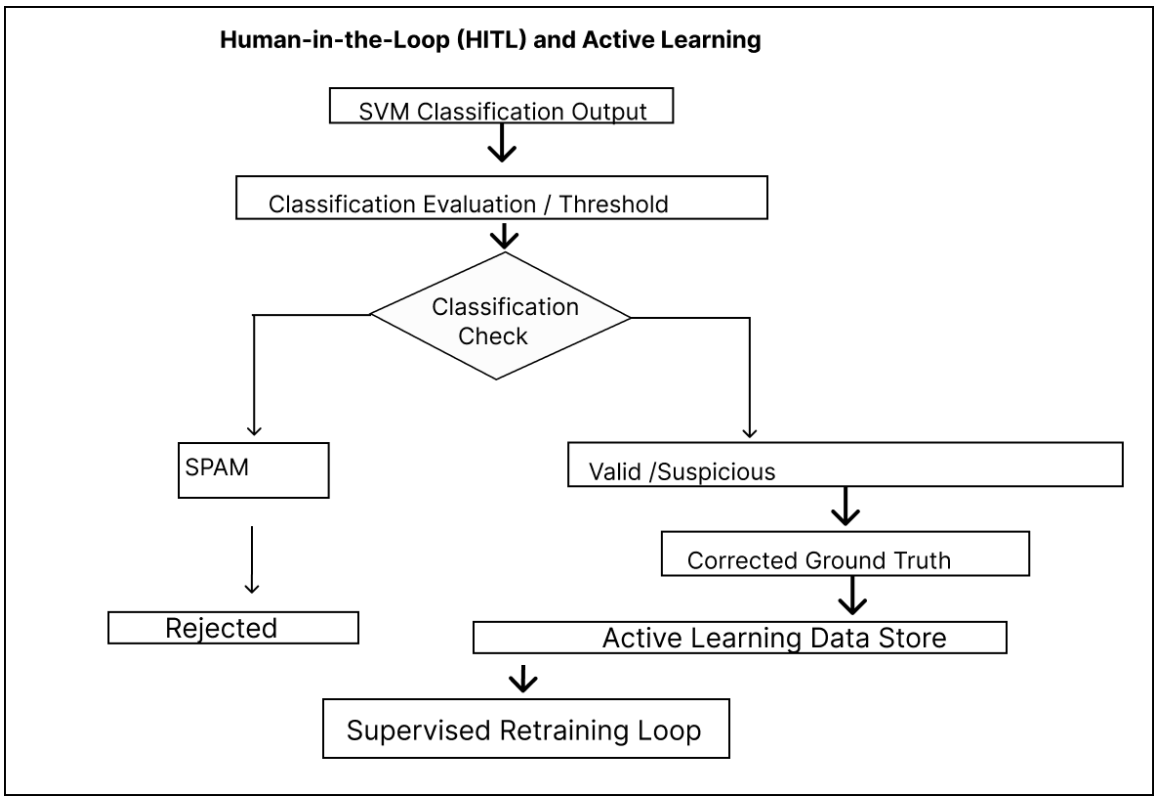
Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification tasks. It works by identifying an optimal hyperplane that separates different classes in a high-dimensional feature space. SVM is well-suited for text classification because of its effectiveness with high-dimensional and sparse datasets, which are common in NLP applications. Studies by Bojjireddy et al. (2021) and Lin (2022) confirm the strong performance of SVM in text classification tasks. In this study, SVM operates on TF-IDF-generated features to classify complaints into valid, suspicious, or spam categories.



2.5.5 Human-in-the-Loop (HITL) and Active Learning

The Human-in-the-Loop (HITL) framework integrates human expertise into the machine learning lifecycle to improve model performance and reliability. It is commonly implemented through Active Learning, a machine learning approach that allows the model to selectively query human experts for labels on uncertain or ambiguous data, thereby improving accuracy while reducing the need for large labeled datasets.

According to Settles (2009), Active Learning is particularly effective in dynamic environments where data patterns continuously evolve, such as changes in language use or emerging spam behaviors. In this study, when barangay administrators correct system-generated classifications, these corrections are recorded as ground truth data and used to retrain the model. This feedback loop enables continuous learning and allows the system to adapt to local linguistic variations and behavioral patterns over time.



2.5.6 State Transition Theory

State transition theory provides the foundation for workflow management within the system. It models a process as a finite set of states and defines the transitions between them based on specific conditions or actions. In the context of this study, state transition theory governs the complaint lifecycle, defining the movement of a complaint through Pending, In Progress, Resolved, and Closed states. This ensures that every complaint is systematically tracked and that barangay officials can monitor progress at each stage, promoting accountability, transparency, and proper resolution.

Chapter 3

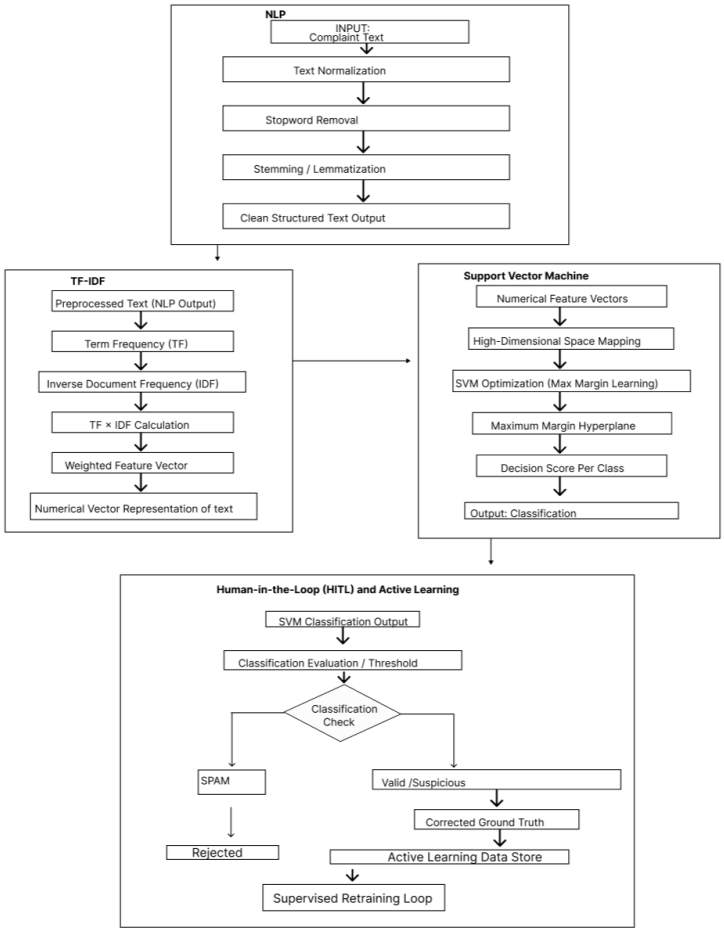
Methodology

3.1 Conceptual Framework

The conceptual framework of this study illustrates how the identified theories are integrated into a unified system design. It shows the relationships between the key components – NLP preprocessing, TF-IDF feature extraction, SVM classification, HITL feedback, and state-transition workflow – as they function together in the proposed Barangay Citizen Complaint System.

The framework operates in two interconnected layers. The first is the classification layer, where raw complaint text submitted in Bisaya, Filipino, or English is preprocessed using NLP techniques, converted into numerical features using TF-IDF, and classified by the SVM model into valid, suspicious, or spam categories. The second is the adaptive and workflow layer, where the HITL mechanism captures administrative corrections to retrain the model over time, and the state-transition workflow tracks each valid complaint through its lifecycle from Pending to Closed.

Together, these components form a closed-loop intelligent system that not only classifies complaints automatically but also improves its accuracy through continuous feedback and ensures structured management of complaint resolution.



3.2 Descriptive Analysis of the Dataset

Before model training and evaluation, the complaint dataset will undergo descriptive and exploratory analysis to establish its composition, quality, representativeness, and possible sources of bias. The study uses textual barangay or public-service complaint narratives and applies Natural Language Processing (NLP), Term Frequency–Inverse Document Frequency (TF-IDF), and Support Vector Machine (SVM) classification. The classification task distinguishes three research labels: Valid Complaint, Suspicious Complaint, and Spam Complaint. The system is designed for multilingual complaint text in Bisaya, Filipino, English, and code-switched combinations.

The raw dataset may consist of de-identified archived complaint or incident records obtained from participating local government or public-service offices, subject to approval and availability. If simulated complaints are needed to supplement gaps in the collected records, they will be clearly marked as simulated and reported separately from real archived records. The primary machine-learning input will be the complaint narrative. Existing office fields such as category, status, resolution, and remarks may be retained for descriptive or annotation context, but they will not automatically be used as predictive SVM features because they may reveal the administrative outcome and create target leakage.

3.2.1 Dataset Composition and Descriptive Reporting

The final descriptive analysis will report the total number of records and the number and percentage of records by data source, research classification, and language. It will also report basic text characteristics such as average and median complaint length, range of word counts, duplicate frequency, missing or unusable narratives, and the proportion of real versus simulated examples. Exploratory data analysis is performed before modeling because understanding data types, distributions, and unusual patterns helps identify problems that may otherwise affect later analysis (Chicco, Oneto, & Tavazzi, 2022).

Table 1: Planned dataset summary to be completed after final annotation

Measure	**Final Count**	**Percentage / Note**
Total complaint records	Pending	100%
Real/archived records	Pending	Report separately
Simulated records, if any	Pending	Report separately
Valid Complaint	Pending	Determine after annotation
Suspicious Complaint	Pending	Determine after annotation
Spam Complaint	Pending	Determine after annotation
Bisaya	Pending	Include code-switched review
Filipino	Pending	Include code-switched review
English	Pending	Include code-switched review
Mixed / code-switched	Pending	Report separately where identifiable

3.2.2 Inherent Biases and Sampling Gaps

The raw complaint source may contain several forms of sampling and representation bias. These biases will be described before model training rather than assumed to have been removed by preprocessing. Baer (2019) notes that biased data may arise from subjective human judgments, real-world behavior already affected by bias, sampling decisions that omit parts of the population, and inappropriate data processing. In complaint records, these concerns are relevant because the archive represents only complaints that were actually submitted, recorded, retained, and made available for research.

- **Source and coverage bias.** Records from a limited number of barangays, offices, time periods, or reporting channels may not represent all complaint behavior in Davao City. Residents who do not use formal complaint channels, complaints that were not digitized, or records that cannot be released may be systematically absent.
- **Class imbalance.** The natural frequency of Valid, Suspicious, and Spam complaints may be unequal. A majority class can dominate training and cause weaker recognition of minority classes. Bao and Barzilay (2022) show that label imbalance

and under-represented groups can contribute to poor generalization.

- **Language representation bias.** Bisaya, Filipino, English, and code-switched complaints may occur at different rates. An under-represented language or code-switching pattern may receive poorer classification performance even when overall accuracy appears high.
- **Historical and administrative bias.** Archived records may reflect earlier human judgments about which complaints were accepted, rejected, prioritized, categorized, or resolved. Human decisions recorded in historical data can be mirrored by later algorithms rather than automatically corrected by them (Baer, 2019).
- **Annotation bias.** The Valid, Suspicious, and Spam labels are research labels and may not already exist in the source office records. Researcher judgment is therefore a potential source of bias, particularly for borderline complaints.
- **Temporal bias and language drift.** Vocabulary, local expressions, code-switching patterns, and spam behavior can change over time, so older archived records may not fully represent future complaint submissions.
- **Simulation bias.** If simulated records are used, they may be cleaner, more regular, or less linguistically diverse than naturally submitted complaints. Simulated examples will therefore be tracked separately and will not be presented as equivalent to real archived data.

Because the final annotated dataset has not yet been completed, it is not methodologically valid to state that Valid, Suspicious, or Spam is already the majority or minority class. After annotation, counts and percentages will be computed for all three classes. The thesis will explicitly identify which class or classes are over-represented or under-represented based on those observed frequencies.

3.2.3 Research Annotation and Control of Human Labeling Bias

Because the source offices are not required to create the research labels, the ground-truth classification will be prepared by the researchers, Lawrence B. Gonzaga and Warren Dominic S. Badong, using a written annotation rubric. The rubric will define operational criteria and examples for Valid, Suspicious, and Spam complaints before full-scale labeling begins.

To reduce inconsistent labeling, both researchers will independently annotate an initial calibration subset. Agreement will be reviewed, and ambiguous cases will be discussed using the rubric. Inter-annotator agreement may be quantified using Cohen's kappa, while disagreements will be resolved through documented consensus. Where a case cannot be resolved confidently, a suitable domain reviewer may be consulted. This procedure helps separate model error from disagreement in the human labels used as training targets.

3.2.4 Data Cleaning and NLP Preprocessing

Data cleaning will be performed before feature extraction. Chicco et al. (2022) emphasize that cleaning and feature engineering are fundamental preprocessing steps and recommend documenting what the data originally looked like and what changes were made for transparency and reproducibility. The study will therefore maintain a raw, unchanged copy of the received de-identified data and a separate processed dataset used for modeling.

The preprocessing pipeline will include the following controls:

- **De-identification verification:** direct personal identifiers that remain in free-text narratives will be removed or masked before modeling.
- **Removal of unusable records:** blank complaint narratives, corrupted rows, and records with no analyzable text will be excluded and counted.
- **Duplicate and near-duplicate review:** exact duplicates will be removed from the modeling dataset so repeated copies of the same complaint do not artificially increase performance. Near-duplicates will be checked to prevent the same complaint from appearing in both training and test data.
- **Text normalization:** unnecessary whitespace and inconsistent encoding will be standardized. Case normalization will be applied consistently while preserving information needed for local-language interpretation.
- **Tokenization and stopword handling:** complaint text will be tokenized. Stopword removal will be applied cautiously using language-appropriate lists so that negation and meaningful local words are not removed without validation.
- **Optional stemming or lemmatization:** these transformations will be retained only if pilot evaluation shows that they improve or maintain validation performance without damaging Bisaya, Filipino, English, or code-switched terms.
- **TF-IDF feature extraction:** only the complaint narrative will be transformed into TF-IDF vectors for the primary classifier. The vectorizer will be fitted on training data only to prevent information from the test set from entering feature construction.

3.2.5 Bias Control, Class Balancing, and Data Curation

Balancing will be decided only after the actual class distribution is known. The validation and test data will retain their observed distributions and will not be artificially oversampled. Any balancing method will be applied only to the training portion of the data or within the training folds of cross-validation.

The primary imbalance control for the SVM will be class weighting, in which minority classes receive greater weight during optimization. If the imbalance remains substantial and minority-class recall is poor, controlled random oversampling of minority training examples may be evaluated. Data curation will also preserve linguistically diverse examples instead of balancing the classes by indiscriminately deleting rare Bisaya or code-switched records.

SMOTE is not selected as the default method for this study. The proposed classifier uses sparse, high-dimensional TF-IDF vectors, and synthetic interpolation in that feature space may create artificial vectors that do not correspond to naturally occurring complaint text. If SMOTE is experimentally compared, it will be applied only inside the training data or training folds and will be retained only if it improves validation performance, particularly macro-F1 and minority-class recall, without degrading held-out performance.

Table 2: Planned bias and control measures

Potential Issue	**Control Method**	**Evaluation**
Class imbalance	Class weighting; optional training-only random oversampling	Per-class precision, recall, F1; macro-F1
Language imbalance	Preserve minority-language examples; stratified review by language where feasible	Performance by language/subgroup
Researcher label bias	Written rubric; dual annotation calibration; adjudication	Agreement rate / Cohen's kappa
Duplicate leakage	Remove duplicates before splitting; group near-duplicates together	Duplicate audit
Administrative target leakage	Exclude status, resolution, and remarks from primary predictive features	Feature audit
Sampling gaps	Document source, office, time, and language coverage	Descriptive counts and limitations

3.2.6 Data Splitting and Control of Overfitting

Overfitting occurs when a model performs very well on training examples but fails to generalize to unseen data. Reitermanová (2010) emphasizes that data splitting and cross-validation are central to estimating generalization and reducing over-training. The study will therefore keep the final test set isolated from preprocessing decisions, balancing, and hyperparameter selection.

A stratified 80/20 train-test split is planned so the three research labels are represented proportionally in both partitions. Hyperparameters will be selected using stratified 5-fold cross-validation within the 80% training portion. The final 20% test set will be used only after the preprocessing choices and model settings have been selected.

For the SVM classifier, overfitting will be controlled primarily through regularization. The SVM regularization parameter C will be tuned during cross-validation to balance training error against model complexity. TF-IDF settings such as minimum document frequency, maximum document frequency, n-gram range, and, if needed, maximum feature count will also be tuned on the training data so that extremely rare tokens or excessive feature growth do not encourage memorization of individual complaints.

Dropout layers are not applicable because the proposed model is an SVM rather than a neural network. Likewise, neural-network early stopping is not the primary control

used in this study. Cross-validation, regularization, feature control, and an untouched held-out test set are the appropriate overfitting controls for the TF-IDF + SVM pipeline. Model selection will consider macro-F1 and per-class precision and recall in addition to accuracy so that good performance on a majority class does not conceal poor performance on a minority class.

3.2.7 Human-in-the-Loop Feedback and Post-Deployment Bias Control

After deployment or controlled system testing, the SVM output is treated as an initial machine prediction rather than the final administrative judgment. Barangay administrators can review and correct misclassified complaints through the Human-in-the-Loop mechanism. Confirmed corrections can be stored as new labeled examples for periodic retraining. However, feedback data will also be monitored for class and language distribution so that repeated corrections from only one office, class, or language do not unintentionally shift the model toward a narrow subset of users.

3.3 Operational Framework

The operational framework describes the end-to-end process of the proposed Barangay Citizen Complaint System, illustrating how citizen complaints are submitted, processed, classified, and managed until resolution. The system operates through a pipeline that combines Natural Language Processing (NLP), machine learning classification, Human-in-the-Loop (HITL) feedback, and workflow management.

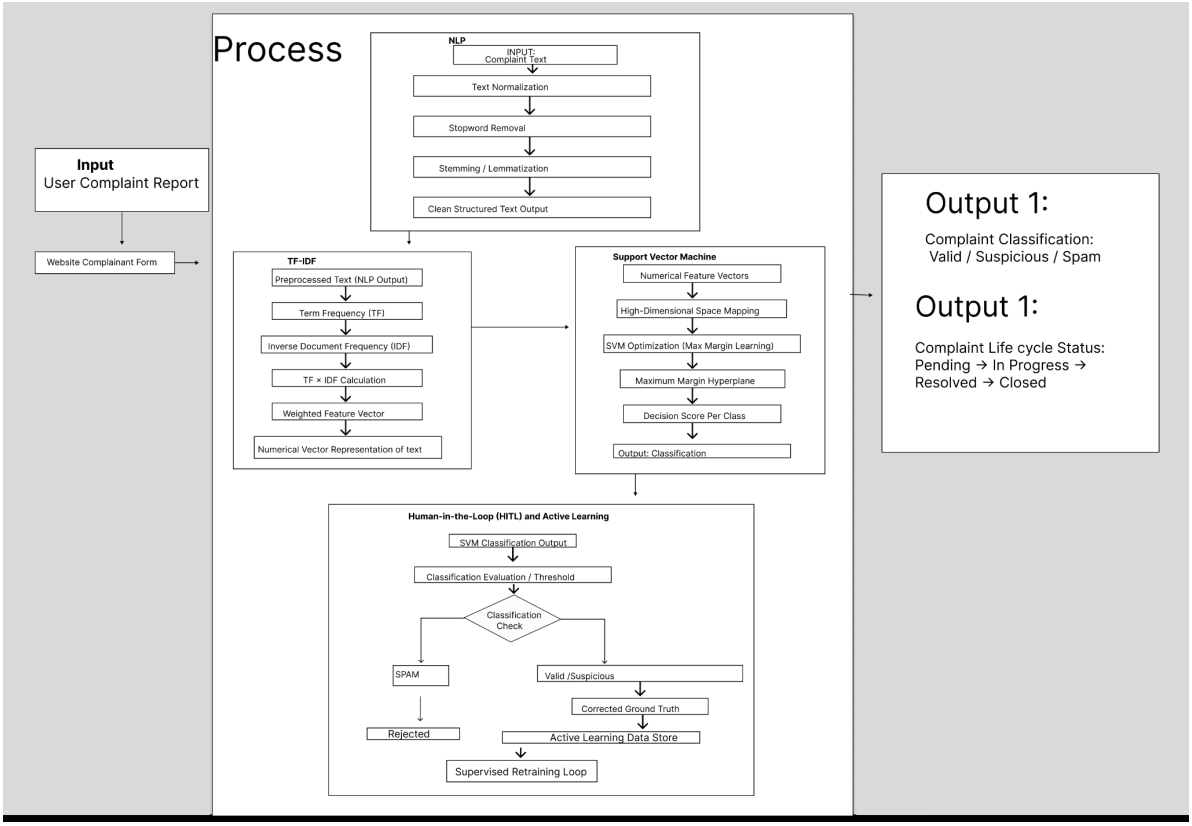
The process begins when citizens submit complaints through the system interface using Bisaya, Filipino, or English. Upon submission, the system captures the complaint text for analysis. The text contains the narrative of the report provided by the user.

Once received, the complaint text undergoes preprocessing to prepare it for analysis. This includes standard Natural Language Processing (NLP) techniques such as text cleaning, tokenization, and stopword removal. The processed text is then transformed into numerical feature representations using Term Frequency–Inverse Document Frequency (TF-IDF), allowing the machine learning model to interpret the text in a structured format. These features are then passed to a Support Vector Machine (SVM) classifier, which analyzes linguistic patterns and assigns an initial classification of Valid, Suspicious, or Spam.

After classification, the results are forwarded to the Barangay Administrative Dashboard, where officials review the system's output. Through the Human-in-the-Loop (HITL) mechanism, administrators are able to correct misclassified complaints. These corrections are stored and used as feedback data for periodic model retraining, allowing the system to continuously improve its accuracy and adapt to evolving language patterns.

Finally, complaints classified as Valid are managed through a structured workflow based on state transition theory. Each complaint follows a lifecycle consisting of Pending, In Progress, Resolved, and Closed states. Barangay officials update the status as actions

are taken, ensuring that every complaint is properly tracked from submission to resolution. This structured workflow enhances transparency, accountability, and efficiency in barangay complaint management.



3.4 Hardware and Software Environment

The machine-learning experiments will be conducted in a fixed and documented environment to support reproducibility. The development computer is equipped with a 13th Gen Intel(R) Core(TM) i7-13700HX processor, 16 GB of installed system memory (15.75 GB usable), an NVIDIA GeForce RTX 4060 Laptop GPU, and Intel(R) UHD Graphics. The 16 GB memory configuration consists of two 8 GB Samsung DDR5-5600 modules. The computer runs Microsoft Windows 11 Home, Version 25H2, OS Build 26200.9457.

The proposed TF-IDF and linear Support Vector Machine workflow is primarily CPU-based; therefore, the CPU will perform the main preprocessing, feature extraction, cross-validation, and classifier training operations. The dedicated GPU is documented as part of the complete hardware environment even though GPU acceleration is not required by the selected scikit-learn implementation.

The experimental software environment is pinned in the project repository to Python 3.13.5, pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0, SciPy 1.17.0, and joblib 1.5.3. These versions are recorded in the repository README and `requirements.txt` so that the same computational environment can be recreated during model development and evaluation.

Table 3: Experimental hardware and software environment

Component	Specification	Purpose
CPU	13th Gen Intel(R) Core(TM) i7-13700HX	Preprocessing, TF-IDF generation, cross-validation, and SVM training
Dedicated GPU	NVIDIA GeForce RTX 4060 Laptop GPU	Documented hardware component; not required for the selected scikit-learn pipeline
Integrated GPU	Intel(R) UHD Graphics	Integrated graphics adapter
RAM	16 GB installed (15.75 GB usable), 2 × 8 GB Samsung DDR5-5600	Stores the dataset, sparse feature matrices, and model-training data
Operating system	Windows 11 Home, Version 25H2, OS Build 26200.9457	Host operating system
Programming language	Python 3.13.5	Primary implementation language
pandas	2.2.3	Data loading, cleaning, filtering, and descriptive analysis
NumPy	2.3.5	Numerical operations
scikit-learn	1.8.0	TF-IDF, classifiers, data splitting, metrics, cross-validation, and grid search
SciPy	1.17.0	Scientific computing and statistical support
joblib	1.5.3	Model persistence and parallel execution support

3.5 Version Control and Reproducibility

Version control and reproducibility are managed through Git and GitHub. The project repository is available at:

`https://github.com/rencze/nlp-barangay-complaint-classification-hitl`

The repository contains the project source code, environment specification, reproducibility instructions, and a minimal working example. The `README.md` describes how to clone the repository, create a Python virtual environment, install the pinned dependencies from `requirements.txt`, and execute the demonstration pipeline. The repository structure includes `src/` for preprocessing, training, and evaluation scripts; `examples/` for a small synthetic demonstration dataset; `data/README.md` for the expected dataset

structure and privacy restrictions; and `models/` and `results/` for generated artifacts.

The minimal working example can be reproduced using the following sequence:

1. Clone the GitHub repository from the URL provided above and open a terminal in the project directory.
2. Create a virtual environment using `py -3.13 -m venv .venv`.
3. Activate the environment and install the dependencies using `pip install -r requirements.txt`.
4. Run `python src/train.py -data examples/demo_complaints.csv`.
5. Re-run the saved-prediction evaluation, when needed, using `python src/evaluate.py -predictions results/test_predictions.csv`.

The file `examples/demo_complaints.csv` is used only to demonstrate that the processing and machine-learning pipeline executes correctly. It is not the research dataset and its outputs will not be reported as thesis findings. The final research experiments will use only the approved, de-identified, cleaned, and annotated complaint dataset described in Section 3.2. If the final source data cannot be redistributed, restricted raw records will remain outside the repository and only the data dictionary, required schema, preprocessing code, and other redistributable materials will be retained.

For reproducibility, randomized procedures will use a documented seed of 42 whenever the underlying method supports randomization. The same held-out test records will be used for the baseline and proposed classifiers, and the Git commit associated with the final experiment will be recorded together with the selected hyperparameters and dependency versions.

3.6 Baseline Selection

The primary machine-learning baseline will be a TF-IDF + Multinomial Naive Bayes (MNB) classifier. Multinomial Naive Bayes is a conventional and computationally efficient text-classification method that has been widely used with frequency-based text representations (McCallum & Nigam, 1998). It is therefore suitable as a standard reference model for determining whether the proposed TF-IDF + Support Vector Machine approach provides a measurable improvement over a simpler probabilistic classifier.

The baseline and proposed classifier will use the same cleaned complaint narratives, the same training partition, and the same held-out test partition. This controls the data used by both models so that performance differences are not caused by different test samples. The baseline configuration will initially use `MultinomialNB(alpha=1.0)`. The primary proposed model will use TF-IDF features with a linear Support Vector Machine implemented through `LinearSVC`. Manual complaint validation will remain an operational comparison for workload reduction but will not serve as the primary statistical machine-learning baseline because it does not generate paired automated predictions on the same test examples.

3.7 Machine-Learning Evaluation Metrics

The classification problem contains three target classes: Valid Complaint, Suspicious Complaint, and Spam Complaint. Because the final class frequencies may be unequal, model performance will not be judged using accuracy alone. Evaluation will be performed on the held-out test set, and class-specific measures will be reported together with aggregate metrics.

3.7.1 Accuracy

Accuracy measures the proportion of test records that are classified correctly:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \tag{3.1}$$

Accuracy provides an overall summary of model performance, but it can be misleading when one class is substantially more frequent than the others.

3.7.2 Precision and Recall

For each class k, precision and recall are defined as:

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \tag{3.2}$$

$$\text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \tag{3.3}$$

where TP_k, FP_k, and FN_k are the true-positive, false-positive, and false-negative counts for class k. Precision indicates how often predictions assigned to a class are correct, while recall indicates how many actual examples of the class are successfully identified. These measures are important because false positives and false negatives may have different consequences in complaint screening.

3.7.3 F1-Score and Macro-F1

For each class k, the F1-score is the harmonic mean of precision and recall:

$$F1_k = 2\left(\frac{\text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}\right). \tag{3.4}$$

Macro-F1 will be used as the primary model-selection metric:

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{K} F1_k, \tag{3.5}$$

where $K = 3$. Macro-F1 gives equal importance to Valid, Suspicious, and Spam regardless of their class frequencies, making it appropriate when class imbalance is possible. Weighted-F1 will also be reported because it summarizes class-level F1-scores while accounting for the number of records in each class.

A confusion matrix will be presented to show how the three classes are confused with one another. Per-class precision, recall, and F1-score will also be reported so that poor recognition of Suspicious or Spam complaints is not hidden by strong performance on a more frequent class.

3.8 Hyperparameter Tuning Strategy

Hyperparameter tuning will use `GridSearchCV` with stratified 5-fold cross-validation inside the 80% training partition defined in Section 3.2.6. The final 20% test partition will remain isolated from feature fitting, balancing decisions, and model selection. Grid search is selected because the proposed search space is limited, explicit, and small enough to evaluate systematically. Macro-F1 will be used as the primary cross-validation scoring metric.

All transformations that learn from the data, including TF-IDF fitting and any training-only balancing procedure, will be performed inside the cross-validation pipeline to prevent validation-fold information from leaking into feature construction or model selection.

Table 4: Planned fixed and variable hyperparameters

Pipeline Component	**Parameter**	**Status**	**Values / Setting**
TF-IDF	`analyzer`	Fixed	`word`
TF-IDF	`norm`	Fixed	`l2`
TF-IDF	`smooth_idf`	Fixed	`True`
TF-IDF	`ngram_range`	Variable	(1,1), (1,2)
TF-IDF	`min_df`	Variable	1, 2, 5
TF-IDF	`max_df`	Variable	0.90, 0.95, 1.00
TF-IDF	`sublinear_tf`	Variable	False, True
Linear SVM	C	Variable	0.1, 1, 10
Linear SVM	`class_weight`	Variable	None, balanced
Linear SVM	`max_iter`	Fixed	5000
Cross-validation	Number of folds	Fixed	5 stratified folds
Randomization	Seed	Fixed	42 where applicable

The best configuration will be selected according to mean cross-validated Macro-F1. After selection, the chosen TF-IDF and SVM configuration will be retrained on the complete 80% training partition and evaluated once on the locked 20% test set. Final test results will not be used to revise preprocessing decisions or hyperparameters.

3.9 Statistical Model Comparison

The study will compare two automated classifiers on the same held-out test dataset: (1) the TF-IDF + Multinomial Naive Bayes baseline and (2) the proposed TF-IDF + linear Support Vector Machine. Because both models generate predictions for the same complaint records, their outcomes are paired. McNemar's test is therefore selected as the primary statistical test for comparing their classification errors (Dietterich, 1998).

Let b represent the number of test complaints classified correctly by the baseline but incorrectly by the proposed SVM, and let c represent the number classified incorrectly by the baseline but correctly by the proposed SVM. The continuity-corrected McNemar statistic is:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}. \tag{3.6}$$

The null hypothesis states that the two classifiers have the same probability of error on the paired test examples. The test will use a significance level of $\alpha = 0.05$ with one degree of freedom. When the number of discordant pairs $(b + c)$ is small, the exact two-sided binomial form of McNemar's test will be used instead of the asymptotic chi-square approximation.

A statistically significant result will indicate that the paired error behavior of the two classifiers differs. Statistical significance will be interpreted together with Macro-F1, per-class recall, and the confusion matrix so that statistical differences are not treated as sufficient evidence of practical improvement by themselves.

If the experimental design is later expanded to three or more classifiers evaluated repeatedly across multiple comparable datasets or resampling partitions, a multi-model non-parametric procedure such as the Friedman test may be considered. Under the current two-model design evaluated on one common held-out test set, McNemar's test is the direct paired comparison.

3.10 Experimental Procedure

The machine-learning experiment will be conducted using the following procedure:

1. Acquire, de-identify, clean, and annotate the complaint dataset using the procedures defined in Section 3.2.
2. Remove duplicate leakage and create a stratified 80/20 train-test split. The test partition will then be locked and excluded from model-selection decisions.
3. Train the TF-IDF + Multinomial Naive Bayes baseline using only the training data.
4. Run `GridSearchCV` for the TF-IDF + `LinearSVC` pipeline using stratified 5-fold cross-validation and Macro-F1 as the scoring metric.
5. Retrain the selected SVM configuration on the complete 80% training partition.

6. Evaluate the baseline and proposed classifier on the same 20% held-out test records.
7. Report accuracy, per-class precision, per-class recall, per-class F1-score, Macro-F1, Weighted-F1, and the confusion matrix.
8. Construct the paired correctness table and perform McNemar's test at $\alpha = 0.05$.
9. Record the software versions, random seed, selected hyperparameters, repository revision, and Git commit associated with the final reported experiment.

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